

# Memory-Aware Feature Design Dominates Quantum-Inspired Architecture in Physics-Informed Solvers for Time-Fractional Integro-Differential Equations

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**Abstract**—Physics-informed neural networks (PINNs) for time-fractional integro-differential equations must approximate a non-local Caputo derivative and a weakly singular Volterra memory term, and their accuracy is sensitive both to the numerical treatment of those operators and to the representation given to the network. We benchmark a classical PINN, a trainable-embedding quantum-inspired (TE) surrogate, and analytic memory-aware input features on a manufactured fractional integro-differential problem, under a fairness-controlled protocol with matched parameter budgets, matched collocation points, and ten seeds per variant. A classical-plus-memory control isolates feature-level gains from architecture-level gains. Analytic memory features are the dominant effect: they reduce final relative  $L_2$  error by  $8.66\times$  over the classical baseline (10/10 seeds,  $p = 0.0020$ ). The TE surrogate also helps independently of features ( $1.96\times$  over the classical baseline, 10/10,  $p = 0.0020$ ), and the combined TE-plus-memory model is the strongest variant overall ( $0.00262 \pm 0.00081$ ). Its margin over the classical-plus-memory control is  $1.58\times$  (9/10,  $p = 0.0488$ ) at the primary operating point, which does not clear a Bonferroni threshold for five comparisons, and  $1.64\times$  (9/10,  $p = 0.0098$ ) at an altered fractional setting  $(\alpha, \beta) = (0.7, 0.3)$ . A post-quantum LayerNorm variant is indistinguishable from the plain TE surrogate ( $p = 0.4922$ ). We report a zero-predictor reference ( $u \equiv 0$ , relative  $L_2 = 1$ ) alongside every table, and we make no claim of quantum advantage: the TE model is a classical surrogate whose sinusoidal feature map has learned, input-dependent frequencies.

**Index Terms**—Physics-informed neural networks, fractional integro-differential equations, Caputo derivative, product integration, graded meshes, quantum-inspired neural networks, memory-aware features.

## I. INTRODUCTION

**T**IME-FRACTIONAL integro-differential systems are a demanding test for neural solvers. Their operators are nonlocal in time, so the state at  $t$  depends on accumulated history rather than on local derivatives alone, and the kernels involved are weakly singular. Automatic differentiation cannot express a Caputo derivative directly, so PINNs for such problems must embed a numerical history operator inside the residual [1], [2].

Two directions have been proposed for improving on a baseline PINN in this setting: architecture-level changes, of which trainable-embedding quantum physics-informed neural networks (TE-QPINNs) are a recent instance [3], [4], and

feature-level priors that encode fractional-memory structure directly in the network input. These are frequently conflated. A quantum-inspired architecture evaluated against a plain baseline can appear to succeed when the gain in fact comes from an accompanying change of input representation.

This letter separates the two. We evaluate four variants against a classical baseline under one residual construction, matched parameter counts, matched collocation budgets, and ten seeds, and we add a *classical-plus-memory* control whose only difference from the baseline is the input representation. This control is what makes the attribution possible, and it is absent from most quantum-inspired PINN comparisons.

Our contributions are: (i) a fairness-controlled benchmark that separates feature-level from architecture-level gains on a fractional integro-differential problem; (ii) evidence that analytic memory features are the dominant effect, larger than any architectural change we tested; (iii) evidence that the TE surrogate contributes an independent and smaller gain, with the combined model strongest overall; and (iv) a negative result for post-quantum LayerNorm stabilization. We report all comparisons against an explicit zero-predictor reference, which we argue should be standard practice in this literature.

## II. PROBLEM AND RESIDUAL CONSTRUCTION

### A. Governing Equation

We solve the time-fractional integro-differential equation

$$D_t^\alpha u(x, t) - (x^2 + 1)u_{xx}(x, t) + \int_0^t \sin(x)(t-s)^{-\beta} u(x, s) ds = f(x, t) \quad (1)$$

on  $x \in [0, 1]$ ,  $t \in (0, 1]$ , with  $u(0, t) = t^\alpha$ ,  $u(1, t) = -t^\alpha$  and  $u(x, 0) = 0$ . Here  $D_t^\alpha$  is the Caputo derivative

$$D_t^\alpha u(x, t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha} \partial_s u(x, s) ds, \quad (2)$$

and the Volterra kernel  $(t-s)^{-\beta}$  is weakly singular but integrable for  $0 < \beta < 1$ . The manufactured solution is  $u_{\text{exact}}(x, t) = t^\alpha \cos(\pi x)$ , with  $f$  derived analytically under the full operator, so that final  $L_2$  and  $L_\infty$  errors are computable exactly and identically across variants.

### B. Discretization

The Caputo term uses an L1 approximation on a graded temporal mesh  $t_n = T(n/N)^r$  with  $r \geq 1$  [5], [6], which concentrates nodes near  $t = 0$  where the solution’s temporal derivative is singular. The Volterra term uses kernel-aware product integration: on each interval  $[t_j, t_{j+1}]$  the solution is interpolated linearly while the singular kernel moments  $\int \tau^{-\beta} d\tau$  and  $\int \tau^{1-\beta} d\tau$  are integrated analytically [7], [8], giving

$$\mathcal{I}_n[u](x) \approx \sum_{j=0}^n w_{n,j} \hat{u}(x, t_j), \quad (3)$$

with weights  $w_{n,j}$  depending only on the mesh and  $\beta$ . Verified against the exact Beta-function value for  $u = s^\alpha$ , this rule converges at second order (error  $1.19 \times 10^{-4} \rightarrow 1.91 \times 10^{-6}$  for  $N_t = 25 \rightarrow 200$ ), whereas applying Gauss–Legendre quadrature to the singular integrand stagnates near  $1.7 \times 10^{-2}$  regardless of refinement.

Both the L1 stencil and (3) are linear in the history values  $u(x, t_j)$ . Each is therefore a fixed weight matrix applied to one history tensor, which we assemble with a single batched network evaluation rather than one evaluation per history node. This is algebraically exact — residuals and parameter gradients agree with the naive loop to  $1.8 \times 10^{-15}$  and  $1.8 \times 10^{-14}$  respectively — and reduces residual cost by about  $60\times$  per step, which is what makes a ten-seed protocol at a converged budget practical.

The network minimizes  $\mathcal{L} = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}}$  with fixed weights across all variants.

### III. MODEL VARIANTS

#### A. TE-Inspired Surrogate

The TE surrogate is a classical PyTorch model that emulates TE-QPINN design ideas [4] without a quantum circuit. Inputs are affine-rescaled; a trainable embedding network produces  $\phi_i$ , from which angle-style coordinates  $\theta_i = \phi_i(x, t) \tilde{z}_i$  are formed against cycled input coordinates. The feature map is  $[\sin \theta, \cos \theta]$  augmented with pairwise terms  $\sin(\theta_i) \cos(\theta_{i+1})$ , followed by a mixing block and an expectation-style readout with an optional residual branch.

We state plainly what this is: a multilayer perceptron on a sinusoidal feature map whose frequencies are learned and input-dependent, i.e. a learned Fourier-feature model [9]. The pairwise terms are products of independently computed scalars and therefore factorize; they are degree-two interaction features, not entanglement. No quantum hardware or simulator is used in the results below, and no quantum advantage is claimed.

#### B. Analytic Memory Features

The memory-aware variants replace the input  $(x, t)$  with

$$\phi_{\text{mem}}(x, t) = [x, t, t^\alpha, t^{1-\alpha}, t^{1-\beta}, \log(1+t), xt, xt^\alpha], \quad (4)$$

evaluated at  $t_{\text{safe}} = \max(t, \epsilon)$  with  $\epsilon = 10^{-8}$  and normalized by deterministic, analytically derived per-feature bounds rather than batch statistics, so train and evaluation behavior coincide.

TABLE I  
TEN-SEED RESULTS AT 8000 EPOCHS,  $\alpha = \beta = 0.5$ . RUNTIME IS SINGLE-THREAD CPU PER RUN.

| Variant            | $L_2$ (mean $\pm$ std)                  | $L_\infty$ | Par. | s   |
|--------------------|-----------------------------------------|------------|------|-----|
| TE + memory        | <b>0.00262 <math>\pm</math> 0.00081</b> | 0.0075     | 2498 | 96  |
| Classical + memory | 0.00415 $\pm$ 0.00173                   | 0.0107     | 2433 | 50  |
| TE + LayerNorm     | 0.01759 $\pm$ 0.00479                   | 0.0583     | 2338 | 106 |
| TE fixed residual  | 0.01834 $\pm$ 0.00380                   | 0.0606     | 2306 | 95  |
| Classical          | 0.03595 $\pm$ 0.00740                   | 0.0880     | 2241 | 44  |
| Zero predictor     | 1.00000                                 | 1.0000     | —    | —   |

TABLE II  
PAIRED TESTS ON FINAL  $L_2$ , TEN SHARED SEEDS. BONFERRONI THRESHOLD FOR FIVE COMPARISONS IS 0.01.

| Comparison (A vs B)        | Ratio         | Wins  | $p$    | $d$   |
|----------------------------|---------------|-------|--------|-------|
| Classical+mem vs Classical | 8.66 $\times$ | 10/10 | 0.0020 | −4.44 |
| TE+mem vs TE fixed         | 6.99 $\times$ | 10/10 | 0.0020 | −4.41 |
| TE fixed vs Classical      | 1.96 $\times$ | 10/10 | 0.0020 | −1.87 |
| TE+mem vs Classical+mem    | 1.58 $\times$ | 9/10  | 0.0488 | −0.80 |
| TE+LayerNorm vs TE fixed   | 1.04 $\times$ | 7/10  | 0.4922 | −0.18 |

The powers  $t^\alpha$ ,  $t^{1-\alpha}$ ,  $t^{1-\beta}$  are motivated by the regularity theory for time-fractional problems, which predicts  $t^\alpha$ -type behavior near the origin [6]. The same feature builder is shared by the classical and TE tracks, so no implementation asymmetry is introduced.

### IV. EXPERIMENTAL PROTOCOL

All variants share (1), the residual construction, the collocation and boundary sampling, and the loss weights; only representation and architecture change. The core setting is a  $50 \times 50$  discretization with mesh grading  $r = 2$ ,  $\alpha = \beta = 0.5$ , and parameter counts matched within 11% (2241–2498). Every variant is trained for 8000 Adam epochs with cosine warmup, and each is run on the same ten seeds  $\{0, 1, 2, 3, 4, 42, 123, 999, 2024, 2025\}$ . Errors are evaluated on a uniform  $100 \times 100$  grid independent of the graded training mesh, and the reported value is the error of the final model state, not the best observed during training.

We report paired Wilcoxon signed-rank tests on final  $L_2$  over the shared seeds, bootstrap 95% confidence intervals on the mean paired difference, and paired effect sizes. With five comparisons, a Bonferroni threshold is  $0.05/5 = 0.01$ ; we state explicitly which comparisons clear it.

*Zero-predictor reference.* Predicting  $u \equiv 0$  gives relative  $L_2 = 1$  by definition, and  $L_\infty = \max |u_{\text{exact}}| = 1$  here. Any variant at or above that line has not learned the solution, and differences among such variants are not meaningful. We include this row in every table.

### V. RESULTS

#### A. Feature-Level Versus Architecture-Level Gains

Table I gives the primary result and Table II the paired comparisons. The largest single effect is the input representation: classical-plus-memory improves on the classical baseline by  $8.66\times$  on every seed, with  $p = 0.0020$  and

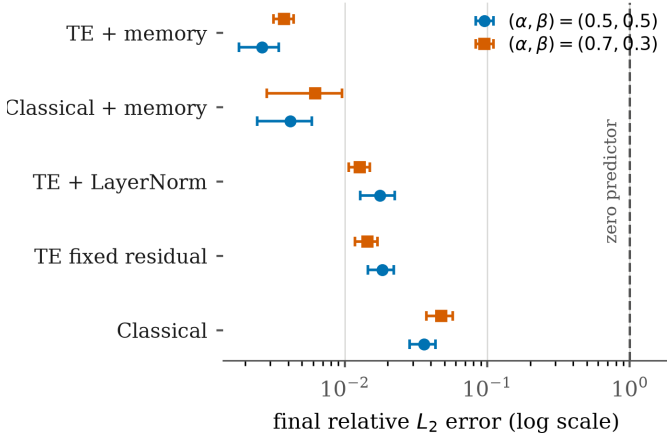


Fig. 1. Final relative  $L_2$  error (mean  $\pm$  std over ten seeds) at both fractional settings. Memory features move a variant roughly an order of magnitude; the TE architecture contributes a smaller additional gain. The ordering is preserved at  $(\alpha, \beta) = (0.7, 0.3)$ .

$d = -4.44$ . The same feature set produces a comparable  $6.99\times$  improvement on the TE track. Both clear the Bonferroni threshold comfortably.

The TE architecture also contributes independently of features: with identical  $(x, t)$  inputs, the TE surrogate improves on the classical baseline by  $1.96\times$  on every seed ( $p = 0.0020$ ). This is a genuine architectural effect, but it is roughly a quarter the size of the feature effect.

The combined TE-plus-memory model is strongest overall. Its margin over the classical-plus-memory control is  $1.58\times$ , winning 9/10 seeds with  $p = 0.0488$ . We do not treat this as established: it does not clear the Bonferroni threshold of 0.01, and the bootstrap interval  $[-0.00266, -0.00041]$ , while excluding zero, is narrow relative to the effect. It is also the more expensive model, at roughly twice the per-epoch cost of classical-plus-memory. We therefore report it as the leading variant rather than as a demonstrated win.

### B. Stabilization

Post-quantum LayerNorm is not supported. Against the plain TE surrogate it gives  $1.04\times$ , wins 7/10, and  $p = 0.4922$ , with a bootstrap interval spanning zero. At a converged budget it is indistinguishable from the variant it was intended to stabilize.

### C. Robustness

Repeating the full ten-seed protocol at  $(\alpha, \beta) = (0.7, 0.3)$  preserves the ordering (Fig. 1): TE+memory  $0.00375 \pm 0.00061$ , classical+memory  $0.00617 \pm 0.00335$ , TE+LayerNorm  $0.01277 \pm 0.00215$ , TE fixed  $0.01439 \pm 0.00258$ , classical  $0.04731 \pm 0.00985$ . Memory features again dominate ( $7.67\times$ , 10/10,  $p = 0.0020$ ) and the TE architecture again contributes independently ( $3.29\times$ , 10/10,  $p = 0.0020$ ). Notably the TE+memory margin over classical+memory is stronger here than at the primary setting ( $1.64\times$ , 9/10,  $p = 0.0098$ ), clearing the threshold that it missed above.

### D. Sensitivity to Training Budget

The conclusions above depend on training to convergence, and we note this because it changes them. Under a severely truncated budget of 12 epochs all five variants score at or above the zero-predictor line (final  $L_2$  between 0.75 and 0.96; four of five with  $L_\infty > 1$ ), and the ordering inverts: the TE surrogate appears *worst* of the group rather than second. Rankings obtained in that regime measure initialization noise. This is a concrete argument for reporting the zero-predictor reference: it makes such a regime immediately visible.

## VI. LIMITATIONS

The memory feature set contains  $t^\alpha$ , which is exactly the temporal factor of the manufactured solution. A linear least-squares fit on the eight features alone reaches relative  $L_2 = 0.124$ , against 0.362 for  $(x, t)$ , so roughly a factor of 2.9 of the measured advantage is available before any network is trained. The trained advantage is  $8.66\times$  and therefore not explained by this alone, but the benchmark is favorable to the feature set, and the effect should be expected to shrink on problems whose time dependence is not  $t^\alpha$ . We regard including  $t^\alpha$  as principled —  $\alpha$  is a known coefficient of (1) and the regularity theory predicts this behavior — but the caveat belongs with the number.

Results are for one problem family, one spatial discretization, and two  $(\alpha, \beta)$  settings. Ten seeds bound the achievable Wilcoxon  $p$  from below at 0.002, which is adequate for the large effects but marginal for the TE-versus-classical-memory comparison. All comparisons are equal-epoch; on an equal-runtime basis the TE variants are penalized by roughly a factor of two.

## VII. CONCLUSION

Under a fairness-controlled ten-seed protocol at a converged budget, analytic memory-aware input features are the dominant source of accuracy improvement for PINNs on a time-fractional integro-differential problem, exceeding any architectural change we tested by roughly a factor of four. A quantum-inspired trainable-embedding surrogate contributes a smaller but real and independent gain, and the combined model is strongest overall, though its margin over the classical-plus-memory control is not established at the primary operating point. Post-quantum LayerNorm stabilization is not supported. We recommend that comparisons of this kind carry both a representation-matched control and an explicit zero-predictor reference; without the first, feature-engineering gains are readily misattributed to architecture, and without the second, rankings can be reported from a regime in which no model has learned the solution.

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vol. 33, 2020, relevant related work: the TE surrogate’s sin/cos feature  
map with learned frequencies is a learned Fourier-feature model.